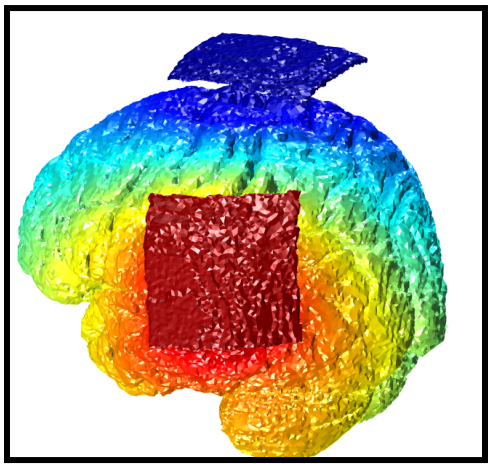
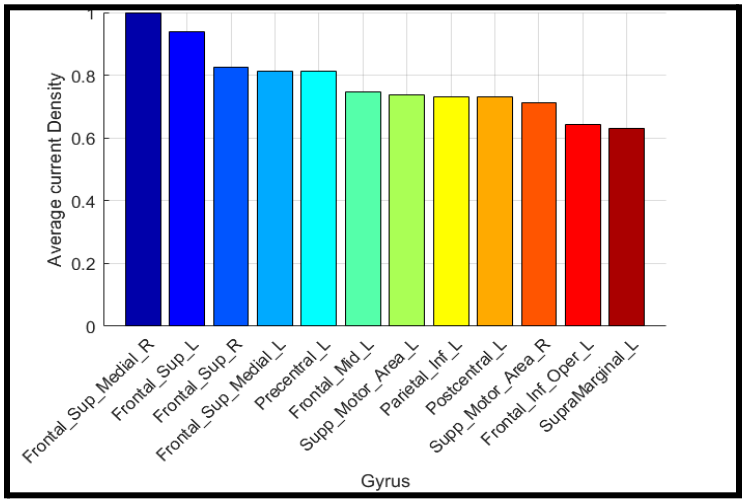
Individual - Systematic Approach For tDCS Analysis (*i*-SATA) -Network

i-SATA Network version 1.0

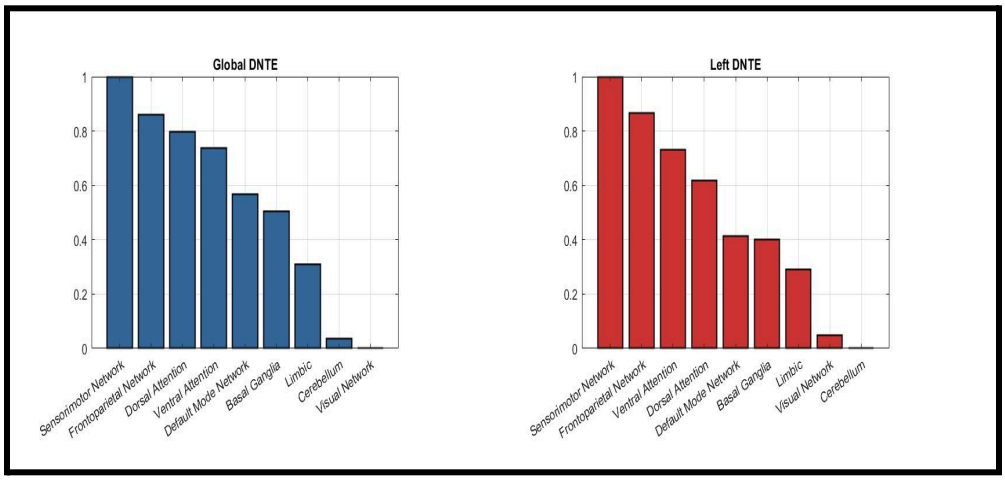
Montage Level Current Distribution



Regional Level Current Distribution



Network Level Current Distribution



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1. Introduction

ᵢSATA stands for Individual Systematic Approach For tDCS Analysis (SATA). It extends SATA (Bhattacharjee et al., 2019) to individual head models. ᵢSATA is essentially a post processing software that acts on the outputs from conventionally used tDCS software like ROAST (Huang, Datta, Bikson, & Parra, 2019). The principles of ᵢSATA are detailed in the paper (Kashyap et al. 2020).

ᵢSATA has been developed using MATLAB vR2019b (The MathWorks Inc., Natick, MA, US). It receives inputs from the software ROAST(v3) and uses acpcdetect v2 (Ardekani et al.1997, 2009), FieldTrip (Oostenveld, Fries, Maris, & Schoffelen, 2011) and SPM12 ("SPM - Statistical Parametric Mapping,") to post process the input data. It uses the brain template from Talairach Client (Lancaster et al.1997, 2000) to superimpose the processed output onto the cortical areas of the brain. The electric field data is overlaid onto the AAL-116 (Automated Anatomical Labeling) atlas. This calculates the specific field intensity received by each of the 116 defined anatomical brain regions. This provides a quantitative index (DTDI - Dose Target Distribution Index) of how well the chosen electrode montage "hits" the intended target. It helps researchers determine if the current is actually reaching the specific ROI (Region of Interest) they intended to stimulate.

ᵢSATA has been updated for network level analysis (iSATA Network). The 116 AAL regions were clustered into 9 Functional Networks (e.g., Default Mode, Frontoparietal, Dorsal Attention, Ventral Attention, Sensorimotor, Visual, Limbic, Basal Ganglia, Cerebellum) as detailed in the work of Hu et al., 2024. The Network Current Density (NwCD) calculates the average intensity received by each functional network. The Dose Network Target Engagement (DNTE) ranks the networks to show which functional system has received maximum stimulation. Since montages are applied to a hemisphere or to an area, we have categorized the DNTE and NwCD specific to each hemisphere (Left, Right, or Combined)

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2. Background

Transcranial Direct Current Stimulation (tDCS) is a technique where a weak current is passed through the electrodes placed on the scalp. The distribution of the electric current induced in the brain due to tDCS is provided by simulation toolbox like Realistic vOlumetric-Approach-based-Simulator-for Transcranial-electric-stimulation (ROAST). However, the procedure to estimate the total current density induced at the target and the intermediary region of the cortex is complex. The Systematic Approach-for-tDCS-Analysis (SATA) was developed to overcome this problem. However, SATA is limited to standardized headspace only. ɨ-SATA overcomes this problem by converting the tDCS simulation results for an individual MRI to match the format of SATA's input. This enables the extension of SATA to individual models as well.

ɨ-SATA provides one pre-processing function that can only be run on Linux - SATA_CBL_ISATA_Preprocess, which is linked to the path folder containing the ROAST simulation for a given individual and a given montage and produces a SATA usable montage - ISATA_Montage.mat in the same folder.

ɨ-SATA also has the 3 major SATA functions (i) average magnitude of current density (MCD) in each cortical lobe, (ii) number of overlapping coordinates, and (iii) cortical areas with high MCD. The measures (i) and (iii) help ascertain the current spread by each montage within a cortical lobe, and (ii) verifies the overlap of the spread of current between a pair of montages. ɨ-SATA specifically aims to facilitate the montage selection process for selectively stimulating a target region of the brain for a given individual. For more details please refer to (Kashyap et al. 2020).

Here we introduce iSATA Network where in the current densities of all areas belonging to a network are averaged to calculate the (i) Network wise Average Current Intensity (NwCD) and (ii) Dose Network Target Engagement (DNTE), which indexes current intensity received by each network (in both hemispheres, and each hemisphere).

The ɨ-SATA toolbox provides a streamlined, automated pipeline to process raw MRI data and generate comprehensive network-level insights on the effects of transcranial Direct Current Stimulation (tDCS). The updated core workflow involves five critical steps:

1. **Individualized Electric Field Simulation:** The process begins by automating ROAST to generate precise 3D electric field maps. This simulation is customized for each subject using their T1-weighted MRI data and the specific electrode configuration employed (e.g., CP5-Cz).
2. **Anatomical Standardization(AC_PC)**: To ensure data comparability, the software performs AC-PC detection and applies standardized reorientation using ART utilities, achieving high anatomical accuracy across subjects.
3. **Regional Dose Quantification**: This step quantifies the Electric Field Distribution at a granular level.
4. The current intensity is calculated for all distinct brain regions defined by the atlas. The effectiveness of the chosen montage is assessed using the Dose Target Determination Index

(DTDI), which confirms if the target Region of Interest (ROI) is being stimulated more effectively than the surrounding cerebral tissue.

5. **This enhancement groups the AAL-116 regions into 9 major functional networks refer to (Hu et al., 2024) (e.g., Default Mode, Frontoparietal, Dorsal Attention, Ventral Attention, Sensorimotor, Visual, Limbic, Basal Ganglia, Cerebellum) to analyze the cognitive impact of tDCS. This functional analysis provides two key metrics:**

- **Network Current Density (NwCD): An index representing the average current intensity received by each of the 9 functional networks. This metric is crucial for understanding which cognitive systems are being modulated by the stimulation.**
- **Dose Network Target Engagement (DNTE): A ranking system that identifies which functional network receives the highest overall dose, with results further segmented by the left and right hemispheres.**

3. Network Division of AAL 116 Regions

Hu et al., 2024 have divided the 116 regions of the AAL atlas into 9 networks (e.g., Default Mode, Frontoparietal, Dorsal Attention, Ventral Attention, Sensorimotor, Visual, Limbic, Basal Ganglia, Cerebellum). Below is the network wise division of the regions provided by them. These networks were further divided into left and right hemisphere to calculate the hemisphere specific NwCD and DNTE.

AAL order	AAL Label	Network name
1	'Precentral_L'	Sensorimotor Network
2	'Precentral_R'	Sensorimotor Network
3	'Frontal_Sup_L'	Dorsal Attention
4	'Frontal_Sup_R'	Dorsal Attention
5	'Frontal_Sup_Orb_L'	Limbic
6	'Frontal_Sup_Orb_R'	Limbic
7	'Frontal_Mid_L'	Frontoparietal Network
8	'Frontal_Mid_R'	Frontoparietal Network
9	'Frontal_Mid_Orb_L'	Frontoparietal Network
10	'Frontal_Mid_Orb_R'	Frontoparietal Network
11	'Frontal_Inf_Oper_L'	Frontoparietal Network
12	'Frontal_Inf_Oper_R'	Frontoparietal Network
13	'Frontal_Inf_Tri_L'	Frontoparietal Network
14	'Frontal_Inf_Tri_R'	Frontoparietal Network
15	'Frontal_Inf_Orb_L'	Default mode Network
16	'Frontal_Inf_Orb_R'	Default mode Network
17	'Rolandic_Oper_L'	Sensorimotor Network
18	'Rolandic_Oper_R'	Sensorimotor Network
19	'Supp_Motor_Area_L'	Sensorimotor Network
20	'Supp_Motor_Area_R'	Sensorimotor Network
21	'Olfactory_L'	Limbic
22	'Olfactory_R'	Limbic
23	'Frontal_Sup_Medial_L'	Default mode Network
24	'Frontal_Sup_Medial_R'	Default mode Network
25	'Frontal_Med_Orb_L'	Default mode Network
26	'Frontal_Med_Orb_R'	Default mode Network
27	'Rectus_L'	Limbic
28	'Rectus_R'	Limbic
29	'Insula_L'	Ventral Attention
30	'Insula_R'	Ventral Attention
31	'Cingulum_Ant_L'	Default mode Network
32	'Cingulum_Ant_R'	Default mode Network
33	'Cingulum_Mid_L'	Ventral Attention
34	'Cingulum_Mid_R'	Ventral Attention
35	'Cingulum_Post_L'	Default mode Network
36	'Cingulum_Post_R'	Default Mode Network
37	'Hippocampus_L'	Limbic
38	'Hippocampus_R'	Limbic
39	'ParaHippocampal_L'	Default Mode Network
40	'ParaHippocampal_R'	Default Mode Network
41	'Amygdala_L'	Limbic
42	'Amygdala_R'	Limbic
43	'Calcarine_L'	Visual Network
44	'Calcarine_R'	Visual Network
45	'Cuneus_L'	Visual Network
46	'Cuneus_R'	Visual Network
47	'Lingual_L'	Visual Network
48	'Lingual_R'	Visual Network
49	'Occipital_Sup_L'	Visual Network
50	'Occipital_Sup_R'	Visual Network
51	'Occipital_Mid_L'	Visual Network
52	'Occipital_Mid_R'	Visual Network
53	'Occipital_Inf_L'	Visual Network
54	'Occipital_Inf_R'	Visual Network
55	'Fusiform_L'	Visual Network
56	'Fusiform_R'	Visual Network
57	'Postcentral_L'	Sensorimotor Network
58	'Postcentral_R'	Sensorimotor Network

59	'Parietal_Sup_L'	Dorsal Attention
60	'Parietal_Sup_R'	Dorsal Attention
61	'Parietal_Inf_L'	Frontoparietal Network
62	'Parietal_Inf_R'	Frontoparietal Network
63	'SupraMarginal_L'	Ventral Attention
64	'SupraMarginal_R'	Ventral Attention
65	'Angular_L'	Default mode Network
66	'Angular_R'	Default mode Network
67	'Precuneus_L'	Default mode Network
68	'Precuneus_R'	Default mode Network
69	'Paracentral_Lobule_L'	Sensorimotor Network
70	'Paracentral_Lobule_R'	Sensorimotor Network
71	'Caudate_L'	basal gangila
72	'Caudate_R'	basal gangila
73	'Putamen_L'	basal gangila
74	'Putamen_R'	basal gangila
75	'Pallidum_L'	basal gangila
76	'Pallidum_R'	basal gangila
77	'Thalamus_L'	basal gangila
78	'Thalamus_R'	basal gangila
79	'Heschl_L'	Sensorimotor Network
80	'Heschl_R'	Sensorimotor Network
81	'Temporal_Sup_L'	Sensorimotor Network
82	'Temporal_Sup_R'	Sensorimotor Network
83	'Temporal_Pole_Sup_L'	Limbic
84	'Temporal_Pole_Sup_R'	Limbic
85	'Temporal_Mid_L'	Default mode Network
86	'Temporal_Mid_R'	Default mode Network
87	'Temporal_Pole_Mid_L'	Limbic
88	'Temporal_Pole_Mid_R'	Limbic
89	'Temporal_Inf_L'	Limbic
90	'Temporal_Inf_R'	Limbic
91	'Cerebelum_Crus1_L'	cerebellum
92	'Cerebelum_Crus1_R'	cerebellum
93	'Cerebelum_Crus2_L'	cerebellum
94	'Cerebelum_Crus2_R'	cerebellum
95	'Cerebelum_3_L'	cerebellum
96	'Cerebelum_3_R'	cerebellum
97	'Cerebelum_4_5_L'	cerebellum
98	'Cerebelum_4_5_R'	cerebellum
99	'Cerebelum_6_L'	cerebellum
100	'Cerebelum_6_R'	cerebellum
101	'Cerebelum_7b_L'	cerebellum
102	'Cerebelum_7b_R'	cerebellum
103	'Cerebelum_8_L'	cerebellum
104	'Cerebelum_8_R'	cerebellum
105	'Cerebelum_9_L'	cerebellum
106	'Cerebelum_9_R'	cerebellum
107	'Cerebelum_10_L'	cerebellum
108	'Cerebelum_10_R'	cerebellum
109	'Vermis_1_2'	cerebellum
110	'Vermis_3'	cerebellum
111	'Vermis_4_5'	cerebellum
112	'Vermis_6'	cerebellum
113	'Vermis_7'	cerebellum
114	'Vermis_8'	cerebellum
115	'Vermis_9'	cerebellum
116	'Vermis_10'	cerebellum

4. Software Download and Installation

3.1 MATLAB version

Important note: For installing *i*-SATA MATLAB version, please refer *i*-SATA user manual section 4 (Kashyap et al. 2020).

In *i*-SATA updated main code (i.e.. iSATA_Network_Ind_Main.m) being used is ROAST v3 (checked for v4). The ROAST can be downloaded from this link: https://www.parralab.org/roast/ For details on how to use ROAST to Simulate tDCS for a given Montage, please refer to the ROAST documentation found at this link: https://github.com/andypotatohy/roast

3.2 Python Version

3.2.1 Prerequisites

1. The Python 3.11 version of *i*-SATA, installation is same as MATLAB version, follow steps under section 3.1 (Note : Errors are likely due to your Python version being too new for the installed MATLAB version.)
2. To ensure dependency stability, either run directly in a specific python version in the terminal (py -3.11version naming before command line) or to use the Conda package manager. Miniforge or Anaconda can be downloaded from https://github.com/conda-forge/miniforge.

3.2.2 Setting up *i*-SATA

The Python version of *i*-SATA utilizes a wrapper to communicate with the ROAST v4 and MATLAB files.

1. Open Powershell and run as administrator
    - cd "C:\Program Files\MATLAB\R2024a\extern\engines\python"
    - py -3.11 setup.py install
2. In powershell run
    - cd D:/ isata_path/ (give isata path in windows)
    - py -3.11 -m pip install -e . (isata will be installed in editable version and '.' at the end to look into current folder)
    - py -3.11 -m pip install nibabel nilearn click numpy scipy matplotlib scikit-learn
    - py -3.11 -m pip install matlabengine==24.1 (Note: Depending on matlab version 24a for R2024b, use ==24.2. and R2023b, use ==23.2)

3. Check installation
    - cd D:/ isata_path/tool
    - py -3.11 isata.py --help

Note : Precede the command with py-3.11. This is recommended even if 3.11 is your default, as it prevents accidental installation into a different Python version. If 3.11 is already your default, you can run commands directly.

4. ɨ-SATA network Usage

4.1 MATLAB

Important note: Before Using ɨ-SATA main code, please refer ɨ-SATA user manual section 4 (Kashyap et al. 2020).

In ɨ-SATA updated main code (i.e.. iSATA_Network_Ind_Main.m) involves 5 core steps. Step 1 to 4 functions remain the same as previous ɨ-SATA version updated the script ease of use (input/output configurations).

4.1.1 Network Analysis

STEP 5 - Network Analysis

The purpose of below command in main script is to analyse network wise current distribution

- [dnte_m, dnte_d] = iSATA_MNI_Network(mni_table, network_path, sub_id, out_path);

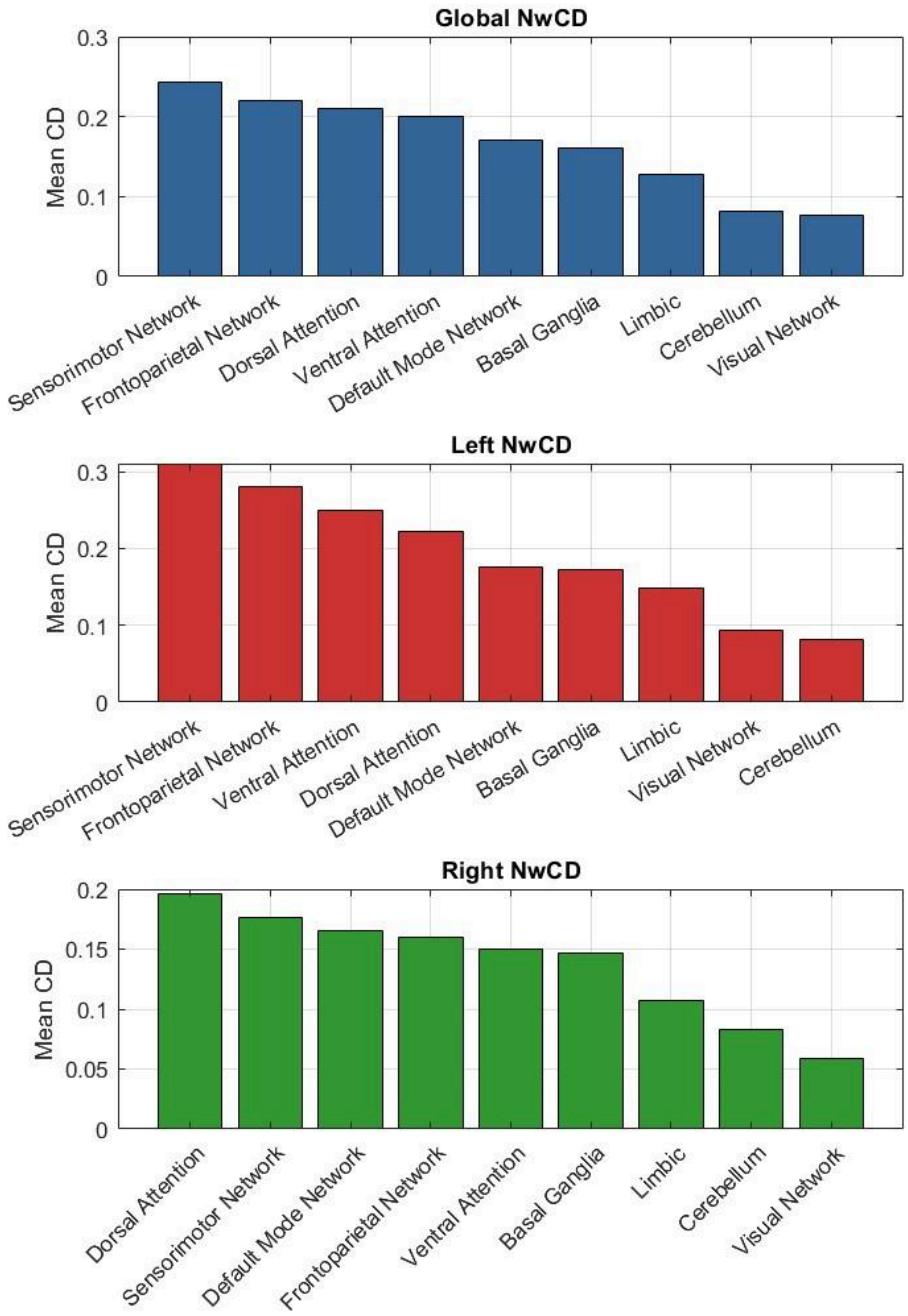
  Grouping the average MNI values of each region for 9 networks to get average current density network wise. In addition to that, the min - max normalization is used (similar to DTDI) to calculate DNTE.

Input :

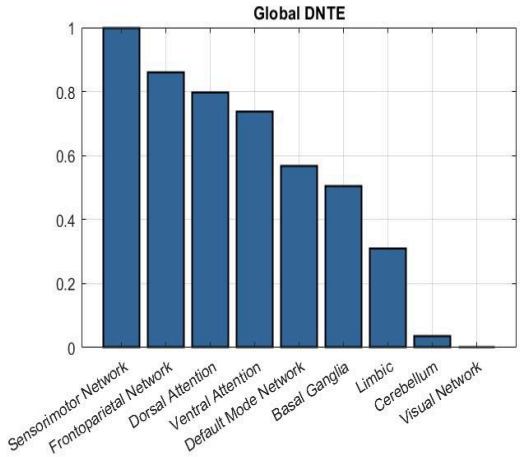
- MNI average current density of each region
- 9 Networks mapped to AAL 116 regions. Please refer to (Hu et al., 2024)

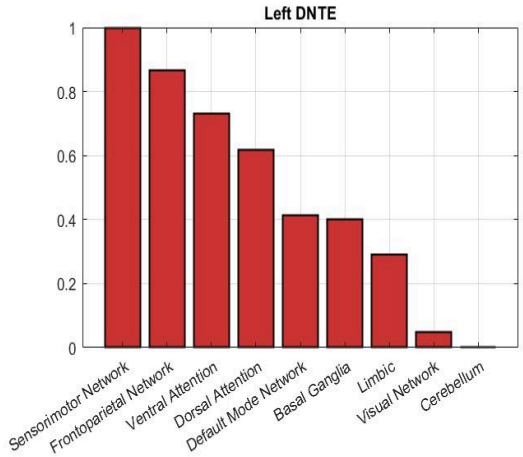
Two Outputs :

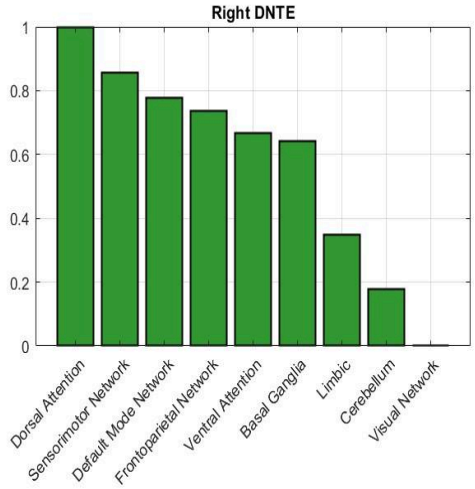
- Network Current Density (NwCD) for Global, Left, and Right networks.

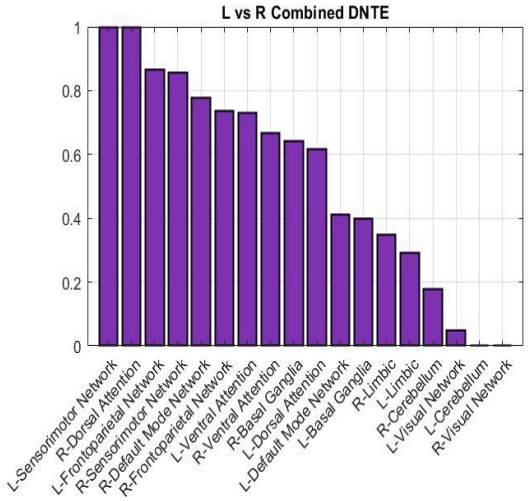


- Dose Network Target Engagement (DNTE)- Will give the rank of current density received by networks and Figure NwCD_DNTE.mat will show DNTE for 9 networks, each hemisphere (L/R) and combined hemispheres (L&R together)









4.2 Python

4.2.1 Terminal Command lines

Step 1 : ROAST

The default ROAST setup for the *i*-SATA Python version is pre-configured; see the table (Table 4.2.1) below for customization options.

Parameter	*i*-SATAFlag	Default	Description
Anode	-a	CP5	Positive electrode location (10-05 system)
Cathode	-k	Cz	Negative/Reference electrode location
Current	-c	2.0	Injected current in milliamperes (mA)
Electrode Type	-t	pad	Shape of the electrode (pad or ring)
Electrode Size	-s	50 50 3	Dimensions in mm (Length, Width, Thickness)
Cap System	-cap	1005	The electrode placement standard (10-05)

Resampling	-r	on	Automatically reslices high-res scans to 1mm
Padding	-p	20	Adds 20mm of zero-space to prevent edge errors.

Table 4.2.1

Spec Feature	ɩ-SATAFlag	Action	Use Case
Multiaxial	-mx	Bypasses standard SPM segmentation	Essential for **atrophied brains** or clinical lesions
Manual GUI	-g	Opens MATLAB window for landmark check	Ensuring **Nasion/Ear landmarks** are correct on first run
Lead Field	-lf	Generates All-Electrode Matrix (L)	Required for **Targeting Optimization**
T2 Support	-t2	Combines T1 and T2 images	Provides **superior skull/CSF segmentation** accuracy

Custom Padding	-p	Increases the 20mm default	For MRI scans that are **tightly cropped** at the neck/crown

Table 4.2.2

For specific feature requirements please add specific *i*SATA flag refer Table 4.2.1, Table 4.2.2 and ROAST V4 features (https://www.parralab.org/roast/)

For example:

Commands shown based on the ISATA\example folder. Here -i followed by input folder path where T1.nii file is present -o followed by output path .

Note : Command line can be used for both individualised and batch processing.

- **Standard**

  py -3.11 isata.py -i "D:\data" -o "D:\results"

- **Frontal Stimulation**

  py -3.11 isata.py -i "D:\data" -o "D:\results" -a F3 -k Fp2 -c 1.5 -s "35 35 3"

- **Padding**

  py -3.11 isata.py -i "D:\data" -o "D:\results" --p 40

- **High-Resolution**

  py -3.11 isata.py -i "D:\data" -o "D:\results" -m fine

- **For Lesioned/Atrophied Brains (Multiaxial Mode):**

  py -3.11 isata.py -i "D:\data" -o "D:\results" -mx

- **To verify Landmark Registration (Manual GUI):**

```
py -3.11 isata.py -i "D:\data" -o "D:\results" -mx -g
```

- **If you have T2 images (Better Segmentation):**

```
py -3.11 isata.py -i "D:\data\sub1_T1.nii" -t2 "D:\data\sub1_T2.nii" -o "D:\results"
```

- **To generate a Lead Field (For later targeting):**

```
py -3.11 isata.py -i "D:\data" -o "D:\results" -lf
```

Step 2 :AC- PC Configuration

- In this input rewritten RAS oriented MRI image gives output AC - PC points in ftOut_ACPC.txt file. (AC - PC coordinates manually also can be saved as ftOut_ACPC.txt file)

```
py -3.11 isata.py acpc -i "D:\results"
```

Step 3 :iSATA MNI

- To get average current density across all regions as MNI and top regions as DTDI

```
py -3.11 isata.py mni-dtdi -i "D:\results"
```

Note : Please refer (Kashyap et al. 2020) to know more details from Step 1 to 4 from the MATLAB version.

Step 4 :iSATA Network

- To get average current density received by each network (AAL-116 atlas is divided into 9 Networks. Please refer to (Hu et al., 2024)) Average Current Density for each Network- is shown as (NwCD) and the Dose Network target engagement (DNTE) provides an index of the current intensity received by each network (in both hemispheres, and each hemisphere). Please refer to the above section (4.1.1 Network analysis) for more details.

```
py -3.11 isata.py dnte -i "D:\results"
```

5. Code Section

5.1 MATLAB

5.1.1 *i*-SATA Network Analysis

```
function [dnte_m, dnte_d] = iSATA_MNI_Network(mni_tab, net_path, sub_id, out_dir)
% iSATA_MNI_Network: Maps current densities to functional networks.
%
% PURPOSE:
%   This function aggregates electric field intensities (current density) from
%   individual AAL regions into large-scale functional brain networks
%   It calculates two distinct metrics:
%   1. NwCD: Raw Network Current Density[cite: 2].
%   2. DNTE: Distributed Network Total Energy (Normalized 0-1)[cite: 2].
%   The function distinguishes between normalization within each hemisphere
%   (Left/Right columns) and normalization against the entire brain's
%   network-hemisphere distribution (Comb_L/Comb_R columns).
%
% INPUTS:
%   mni_tab  - Table containing AAL gyrus names and their average current
%              densities (the 'wt' table from Step 1).
%   net_path - Path to the '.mat' file containing the 1x116 AAL network
%              mapping struct  refer to  (Hu et al., 2024)
%   sub_id   - String identifier for the subject (e.g., 'sub02').
%   out_dir  - Path to the directory where results and figures will be saved.
%
% OUTPUTS:
%   dnte_m   - Table of raw mean intensities (NwCD) for Global, Left, and Right networks[cite: 2].
%   dnte_d   - Table of normalized intensities (DNTE) with combined hemispheric rankings.
%   Files    - Saves '[sub_id]_NwCD_DNTE.fig' (interactive pop-up figure) to out_dir.
  % 1. Load Network Mapping
  data = load(net_path);
  ROI_struct = data.(char(fieldnames(data)));
  network_list = struct2table(ROI_struct);

  % 2. Join MNI table with Network mapping
  combined = innerjoin(mni_tab, network_list, 'Keys', 'Gyrus');

  % 3. Hemisphere Splitting
  gyrus_names = string(combined.Gyrus);
  isLeft  = endsWith(gyrus_names, '_L');
  isRight = endsWith(gyrus_names, '_R');
```

```
% 4. Aggregate Raw Intensities (NwCD)
nw_global = groupsummary(combined, 'Network', 'mean', 'Average_CD');
nw_left   = groupsummary(combined(isLeft,:), 'Network', 'mean', 'Average_CD');
nw_right  = groupsummary(combined(isRight,:), 'Network', 'mean', 'Average_CD');

% --- Step A: DNTE Normalization (0-1) ---
dnte_d = table(nw_global.Network, 'VariableNames', {'Network'});

% Helper function for 0-1 scaling
scale01 = @(x) (x - min(x)) / (max(x) - min(x));

dnte_d.Global = scale01(nw_global.mean_Average_CD);
dnte_d.Left   = scale01(nw_left.mean_Average_CD);
dnte_d.Right  = scale01(nw_right.mean_Average_CD);

% Combined Ranking (L+R pooled)
all_raw_vals = [nw_left.mean_Average_CD; nw_right.mean_Average_CD];
all_norm_vals = scale01(all_raw_vals);

% --- Step B: NwCD Table (Raw Data Storage) ---
dnte_m = table(nw_global.Network, nw_global.mean_Average_CD, ...
            nw_left.mean_Average_CD, nw_right.mean_Average_CD, ...
            'VariableNames', {'Network', 'Global', 'Left', 'Right'});
% --- FIGURE 1: NwCD RAW VALUES (3 VERTICAL PLOTS) ---
fig1 = figure('Color', 'w', 'Name', 'NwCD Analysis', 'Units', 'normalized', 'Position', [0.05 0.05 0.4 0.9]);

% Global NwCD
subplot(3,1,1);
[valG, idxG] = sort(dnte_m.Global, 'descend');
XG = categorical(dnte_m.Network(idxG)); XG = reordercats(XG, string(XG));
bar(XG, valG, 'FaceColor', [0.2 0.4 0.6]);
ylabel('Mean CD'); title('Global NwCD '); grid on;
% Left NwCD
subplot(3,1,2);
[valL, idxL] = sort(dnte_m.Left, 'descend');
XL = categorical(dnte_m.Network(idxL)); XL = reordercats(XL, string(XL));
bar(XL, valL, 'FaceColor', [0.8 0.2 0.2]);
ylabel('Mean CD'); title('Left NwCD '); grid on;
% Right NwCD
subplot(3,1,3);
[valR, idxR] = sort(dnte_m.Right, 'descend');
XR = categorical(dnte_m.Network(idxR)); XR = reordercats(XR, string(XR));
bar(XR, valR, 'FaceColor', [0.2 0.6 0.2]);
ylabel('Mean CD'); title('Right NwCD '); xtickangle(45); grid on;

savefig(fig1, fullfile(out_dir, [sub_id '_NwCD.fig']));
% --- FIGURE 2: DNTE NORMALIZED (4-PANEL GRID) ---
fig2 = figure('Color', 'w', 'Name', 'DNTE Analysis', 'Units', 'normalized', 'Position', [0.5 0.05 0.4 0.9]);
```

```
  % Global DNTE
  subplot(2,2,1);
  [dvalG, didxG] = sort(dnte_d.Global, 'descend');
  DXG = categorical(dnte_d.Network(didxG)); DXG = reordercats(DXG, string(DXG));
  bar(DXG, dvalG, 'FaceColor', [0.2 0.4 0.6], 'LineWidth', 1.2);
  title('Global DNTE'); grid on;
  % Left DNTE
  subplot(2,2,2);
  [dvalL, didxL] = sort(dnte_d.Left, 'descend');
  DXL = categorical(dnte_d.Network(didxL)); DXL = reordercats(DXL, string(DXL));
  bar(DXL, dvalL, 'FaceColor', [0.8 0.2 0.2], 'LineWidth', 1.2);
  title('Left DNTE'); grid on;
  % Right DNTE
  subplot(2,2,3);
  [dvalR, didxR] = sort(dnte_d.Right, 'descend');
  DXR = categorical(dnte_d.Network(didxR)); DXR = reordercats(DXR, string(DXR));
  bar(DXR, dvalR, 'FaceColor', [0.2 0.6 0.2], 'LineWidth', 1.2);
  title('Right DNTE'); xtickangle(45); grid on;
  % Combined L vs R DNTE
  subplot(2,2,4);
  all_labels = [strcat('L-', dnte_d.Network); strcat('R-', dnte_d.Network)];
  [sorted_vals, sort_idx] = sort(all_norm_vals, 'descend');
  DX_Comb = categorical(all_labels(sort_idx)); DX_Comb = reordercats(DX_Comb, string(DX_Comb));
  bar(DX_Comb, sorted_vals, 'FaceColor', [0.5 0.2 0.7], 'LineWidth', 1.2);
  title('L vs R Combined'); xtickangle(45); grid on;

  savefig(fig2, fullfile(out_dir, [sub_id '_DNTE.fig']));
end
```

5.1.2 *i*-SATA Main Script

```
%% STEP 3,4 &5 are combined in below script: ISATA Complete Network Analysis
%
    ==========================================================================
    ===
% PURPOSE:
%   To calculate how much electric current engages specific Brain area
%   The ranking of current intensity according to target selection
% Amount of current density received by each network - 9 networks are there
% The networks are also divided into left and right hemispheres
%% network Names "Default Mode", "Frontoparietal", "Dorsal Attention",
% "Ventral Attention", "Sensorimotor", "Visual", "Limbic",
% "Basal Ganglia", "Cerebellum" Please refer to  (Hu et al., 2024)
% SATA_CBL_Coords_With_High_MCD_MNI_atlas: Maps E-field to AAL atlas regions.
```

```
% INPUTS:
%   Montage   - Structure containing:
%             .Target.FaceVertexCData (E-field values)
%             .Target.Vertices (Mesh coordinates)
%             .ISATA (AC, PC, and MSP coordinates) [cite: 2026-03-21]
%             .sub_id (Subject identifier string)
%   atlas_nii - Path to the AAL atlas NIfTI file (.nii).
%   atlas_mat - Path to the AAL atlas label list (.mat).
%   out_dir   - Path to the directory where results and plots will be saved.

% OUTPUTS:
%   MNI_table - Table of Gyrus names and Average Current Density in
%               original atlas order.
%   DTDI_table     - Table of regions ranked by normalized DTDI values.
%   Files          - Saves '[sub_id]_MNI.mat', '[sub_id]_DTDI.mat', and
%               '[sub_id]_MNI_DTDI.png' to the output directory.
% NwCD - Network current density of all networks
%Figure 1 will show NwCD for 9 networks, each hemisphere (L/R) and combined
%hemispheres (L&R together)
% DNTE- Will give the rank of current density received by networks
%Figure 2 will show DNTE for 9 networks, each hemisphere (L/R) and combined
%hemispheres (L&R together)
%
    ==========================================================================
    ====
clear; clc;
% --- CONFIG ---
sub_id    = 'sub02';
in_path   = 'D:\sample_data\iSATA_results\sub02_T1w\roast';
out_path  = 'D:\sample_data\iSATA_results\sub02_T1w\isata';
% FIX: Use single quotes to ensure this is a character array for SPM
atlas_nii = 'C:\Users\user\Downloads\SATA\Utilities\Atlas\aal_for_SPM12\ROI_MNI_V4.nii';
atlas_mat =
    'C:\Users\user\Downloads\SATA\Utilities\Atlas\aal_for_SPM12\ROI_MNI_V4_List.mat';
if ~exist(out_path, 'dir'); mkdir(out_path); end
% --- BUILD MONTAGE ---
acpc_file = fullfile(in_path, 'ACPC_coordinates.mat');
if exist(acpc_file, 'file'); load(acpc_file); else; error('ACPC file not found!'); end
% ROBUST FILE SEARCH - Prevents "Search term must be text" error
mat_files = dir(fullfile(in_path, '*.mat'));
found_sim = false;
for i = 1:length(mat_files)
  curr_file = mat_files(i).name;
  % Ensure we are checking a valid string
```

```
  if ~isempty(curr_file) && contains(string(curr_file), 'T') && ~contains(string(curr_file), 'ACPC')
      load(fullfile(in_path, curr_file));
      found_sim = true;
      break;
  end
end
if ~found_sim; error('Could not find the simulation T-pattern file!'); end
% --- DATA RETRIEVAL ---
e_pos_list = dir(fullfile(in_path, '*e.pos'));
if isempty(e_pos_list); error('No .pos file found!'); end
% Load ROAST node/elem if not already in workspace
% load(fullfile(in_path, 'your_roast_output.mat'));
[Coordinates, Electric_Field] = SATA_CBL_Retrieve_Coords(e_pos_list(1), node, elem);
% ---  MSP DETECTION (Step 3) ---
% You must detect the MSP before building the montage
MSP = SATA_CBL_Detect_MSP(Coordinates, AC, PC);
% ---  MONTAGE CONSTRUCTION ---
ISATA_Montage = struct();
ISATA_Montage.sub_id = sub_id;
ISATA_Montage.Target.FaceVertexCData = Electric_Field;
ISATA_Montage.Target.Vertices = Coordinates;
ISATA_Montage.ISATA.AC = AC; ISATA_Montage.ISATA.PC = PC;
ISATA_Montage.ISATA.MSP = MSP;
% ---  RUN MAPPING (Step 4) ---
[mni_table, dtdi_table] = SATA_CBL_Coords_With_High_MCD_MNI_atlas(ISATA_Montage,
    atlas_nii, atlas_mat, out_path);
% Save Step 4 Plots as .fig in output directory
set(findobj('Name', 'i-SATA AAL 116 Results'), 'Visible', 'on');
savefig(gcf, fullfile(out_path, [sub_id '_MNI_DTDI.fig']));
% ---  NETWORK ANALYSIS (Step 5) ---
network_path =
    'C:\Users\user\Downloads\SATA\Utilities\Atlas\aal_for_SPM12\aal_MNI_V4_Network.mat';
if exist('mni_table', 'var') && ~isempty(mni_table)
  % Execute network function
  [dnte_m, dnte_d] = iSATA_MNI_Network(mni_table, network_path, sub_id, out_path);

  % Save data tables
  save(fullfile(out_path, [sub_id '_NwCD.mat']), 'dnte_m');
  save(fullfile(out_path, [sub_id '_DNTE.mat']), 'dnte_d');
end
```

5.2 Python

Run all steps for all T1.nii files in 'data' folder

- py -3.11 isata.py run-all -i roast "D:\data" -o "D:\iSATA_Results"

Run separate steps on the output directory

- py -3.11 isata.py acpc -i "D:\iSATA_Results"
- py -3.11 isata.py mni-dtdi -i "D:\iSATA_Results"
- py -3.11 isata.py dnte -i "D:\iSATA_Results"

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- Matlab: https://www.mathworks.com/products/matlab.html
- SPM: https://www.fil.ion.ucl.ac.uk/spm/
- Fieldtrip: http://www.fieldtriptoolbox.org/
- Talairach client: http://www.talairach.org/client.html
- Acpcdetect: https://www.nitrc.org/forum/forum.php?forum_id=1927
- ROAST: https://www.parralab.org/roast/
- **Python:** python.org

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